

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. (EC) Examination

November 2013

EC 301 Electromagnetic Theory

Date: 18.11.2013, Monday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) State the Bio-Savart Law. [02]
(b) State and Prove uniqueness theorem. [05]
- Q - 2 (a) Using Gauss's Law, derive the equation of charge enclosed by differential volume element in terms of Divergence. [07]
(b) A point charge $Q_1 = 10\mu\text{C}$ is located at $P_1 (1, 2, 3)$ in free space, while $Q_2 = -5\mu\text{C}$ is at $P_2 (1, 2, 10)$ (i) Find the vector force exerted on Q_2 by Q_1 (ii) find the coordinates P_3 at which a point charge Q_3 experiences no force. [07]

OR

- Q - 2 (a) Explain ampere's circuital law for an infinitely long coaxial transmission line. [07]
(b) Find the vector magnetic field intensity in Cartesian coordinates at $P(1.5, 2, 3)$ caused by a current filament of 24 A in the $\mathbf{a}_z$ direction on the Z axis and extending from: $z = -\infty$ to $+\infty$ [07]
- Q - 3 (a) Tabulate the maxwell's equations in point form and integral forms for time varying fields. [07]
(b) A non uniform field $\mathbf{E} = y \mathbf{a}_x + x \mathbf{a}_y + 2 \mathbf{a}_z$ is given. Determine the work expended in carrying 2C charge from B(1,0,1) to A(0.8,0.6,1) [07]
(i) Along the shorter arc of circle $x^2 + y^2 = 1, z = 1$.
(ii) Along the straight line path from B to A.

OR

- Q - 3 (a) Derive the magnetic field intensity caused by a infinite length current filament on Z axis. [07]
(b) State the faraday's induction law. Given the magnetic flux density, $\mathbf{B} = 6\cos 10^6 t \sin 0.01x \mathbf{a}_z$ mT. Find (i) the magnetic flux passing through the surface $z = 0, 0 < x < 20 \text{ m}, 0 < y < 3 \text{ m}$ at $t = 1 \mu\text{s}$ (ii) The value of the closed line integral of Electric Field Intensity around the perimeter of surface specified above at $t = 1$ [07]

SECTION – II

- Q - 4 (a) Explain skin effect [02]
(b) Elaborate nature of dielectric materials [03]
- Q - 5 (a) Using neat sketches thoroughly describe Cylindrical Coordinate system. In addition, perform conversion from cylindrical coordinate system to Cartesian coordinate system [06]
(b) Consider the two potential fields $V_1 = x$ and $V_2 = x + \sin x$. Calculate following: [06]
(i) $\nabla^2 V_1$ and $\nabla^2 V_2$ (ii) V_1 and V_2 at $x = 0$ (iii) V_1 and V_2 at $x = \pi$ (iv) Are V_1 and V_2 identical? Justify the answer

OR

- Q - 5 (a) Using neat sketches thoroughly describe Spherical Coordinate system. In addition, perform conversion from spherical coordinate system to Cartesian coordinate system [06]
(b) A loop of wire is constructed of three straight segments connecting (0,0,0) to (0.6,0,0) to (0.4,1,0.7) to (0,0,0). A current of 8 mA is in the $\mathbf{a}_x$ direction in the first segment. Given a uniform magnetic field, $\mathbf{B} = 0.2 \mathbf{a}_x - 0.1 \mathbf{a}_y + 0.2 \mathbf{a}_z$ T. Calculate following: [06]
(i) a force on the segment extending from (0,0,0) to (0.6,0,0). (ii) the total force on the loop

- Q - 6 Attempt Any Three [18]
1. Explain Poynting's Theorem and Wave Power with use of necessary equation and give its significance.
 2. Derive equation for intrinsic impedance in dielectrics
 3. Using sketches describe magnetic boundary conditions
 4. Describe current continuity equation